EXECUTIVE SUMMARY: End-to-End Sales & Churn Analytics

The Business Problem

Our organization faces significant Monthly Recurring Revenue (MRR) loss due to customer churn. Historically, our CRM data (subscriptions) and POS data (hardware sales) have been siloed. This capstone project merged 500 customer records with transactional sales data to uncover cross-database attrition drivers and calculate true Customer Lifetime Value (CLTV).

Key Statistical Findings

1. **The Ecosystem Effect (Hardware Reduces Churn):** Statistical T-Tests ($p < 0.05$) mathematically prove that customers who purchase hardware (Laptops, Phones) in addition to their subscription are significantly less likely to churn than subscription-only customers.
2. **The "Bill Shock" Vulnerability:** Despite the hardware loyalty effect, high MonthlyCharges remain a critical risk factor. A secondary T-Test ($p < 0.05$) confirmed that when billing crosses into the highest percentiles, churn risk spikes universally across all customer segments.

Strategic Implementation Plan

Based on empirical data, we recommend a 3-phase rollout to reduce month-to-month attrition by 20%:

- **Phase 1 (Days 1-30):** Launch an aggressive hardware cross-selling campaign. Subsidize entry-level hardware as a "loss leader" for subscription-only customers to lock them into the ecosystem.
- **Phase 2 (Days 31-60):** Introduce an automated retention program. When a customer's usage pushes their monthly bill above the 75th percentile (0e.g., >$90/mo), automatically trigger a loyalty discount or upgrade them to a fixed-price tier to prevent "bill shock."
- **Phase 3 (Days 60-90):** Shift the core business model from "Subscriptions + Separate Hardware" to bundled packages, permanently forcing the high-retention hardware effect across the entire customer base.